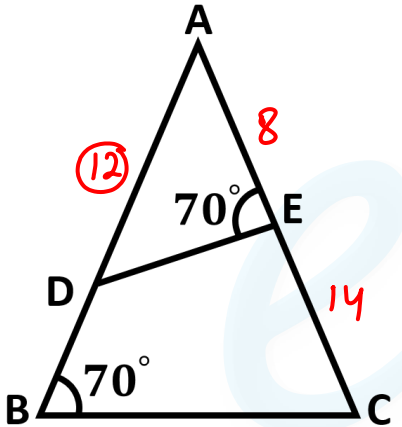


(1) In the given figure, If $AD = 12cm, AE = 8cm$ and $EC = 14cm$, then what is the value in (in cm) of BD?

दी गई आकृति में, यदि $AD = 12cm$, $AE = 8cm$ तथा $EC = 14cm$ है, तो BD का मान (cm में) क्या है?

a) $\frac{50}{3}$ b) 15 c) $\frac{8}{3}$ d) $\frac{44}{3}$

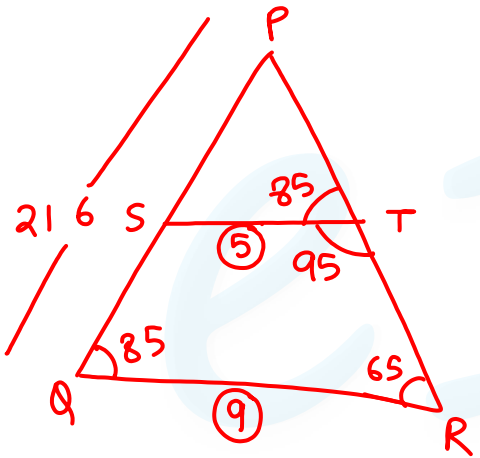
$\frac{3}{12} = \frac{8^2}{AB}$

22

$AB = \frac{44}{3} - 12 = \frac{8}{3}$

coaching center

CGL 2017 T2



$\frac{PT}{21.6} = \frac{5}{9}$

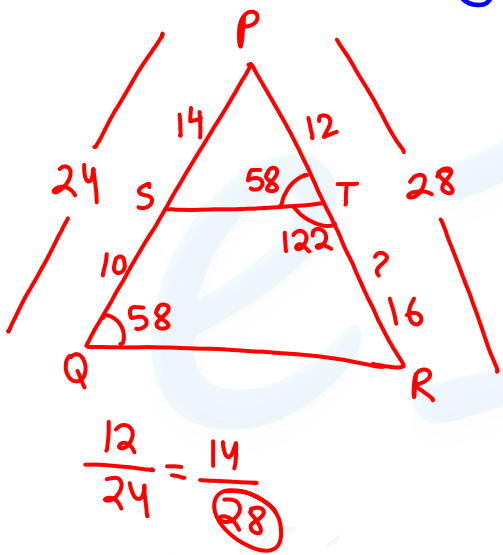
2.4

② In $\Delta PQR, \angle Q = 85°$ and $\angle R = 65°$. Points S and T are on the sides PQ and PR respectively such that $\angle STR = 95°$ and the ratio of QR and ST is $9:5$. If $PQ = 21.6\ cm$, then the length of PT is:

किसी त्रिभुज PQR में $\angle Q = 85°$ और $\angle R = 65°$ है| बिंदु S और T क्रमशः PQ और PR पर स्थित है जहां $\angle STR = 95°$, तथा QR और ST का अनुपात $9:5$ है यदि $PQ = 21.6\ cm$, तो PT की लम्बाई ज्ञात करो?

a) 10.5 cm
b) 9.6 cm
c) 12 cm
d) 9 cm

CGL 2019 T1

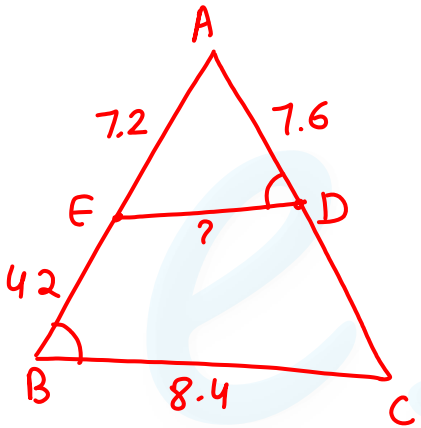


(3) In ΔPQR, PQ= 24 cm and $\angle Q = 58°$. S and T are the points on side PQ and PR respectively, such that $\angle STR = 122°$. If $PS = 14\ cm$ and $PT = 12\ cm$, then the length of RT is :

ΔPQR में, $PQ = 24\ cm$ और $\angle Q = 58°$| S और T क्रमशः भुजा PQ और PR पर स्थित ऐसे बिंदु है कि $\angle STR = 122°$ | यदि $PS = 14\ cm$ और $PT = 12\ cm$, तो RT की लम्बाई ज्ञात कीजिये|

a) 14.8 cm
b) 16 cm
c) 15 cm
d) 16.4 cm

coaching center



$\frac{DE}{8.4} = \frac{7.6}{11.4}$ 4.2

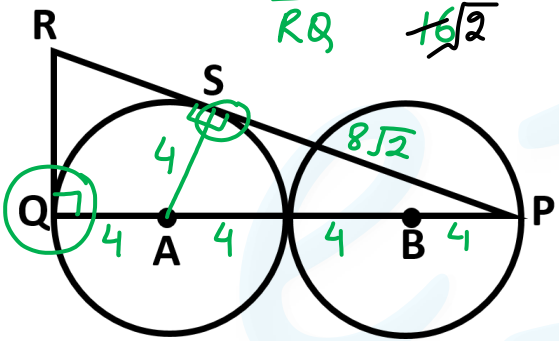
2.8 6

④ In ΔABC, D and E are the points on sides AC and AB respectively such that $\angle ADE = \angle B$. If $AD = 7.6\ cm, AE = 7.2\ cm, BE = 4.2\ cm$ and $BC = 8.4$ cm then DE is equal to :

ΔABC में, D और E क्रमशः AC और AB पर बिंदु है जिससे $\angle ADE = \angle B$, यदि $AD = 7.6$ सेमी, $AE = 7.2$ सेमी, $BE = 4.2$ सेमी, $BC = 8.4$ सेमी, तो DE का मान ज्ञात करें:

a) 6.3 cm
b) 5.6 cm
c) 7.4 cm
~~d)~~ 5.6 cm

CHSL 2018



$\frac{4}{RQ} = \frac{8\sqrt{2}}{16\sqrt{2}}$

14

$\frac{+2}{3}$

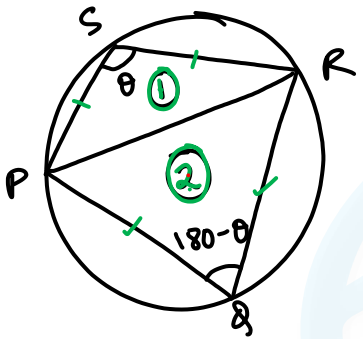
(5) In the given figure, two identical circles of radius 4cm touch each other, A and B are the centres of the two circles, if RQ is a tangent to the circle, then. What is the lenght (in cm) of RQ?

दी गई आकृति में, दो सामान वृत जिनकी त्रिज्या 4 cm है, एक दुसरे को सपर्श कर रहे है| दोनों वृतों के केंद्र A तथा B है. यदि RQ वृत पर एक सपर्शरेखा है, तो RQ की लम्बाई (cm में) क्या है?

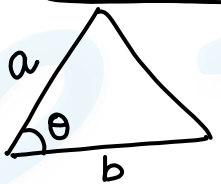
a) $3\sqrt{3}$ b) $2\sqrt{6}$

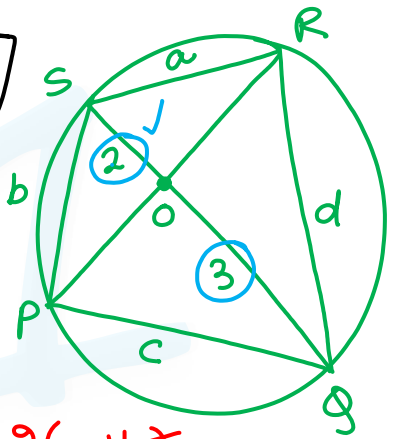
c) $4\sqrt{2}$ d) $\frac{16}{3}$

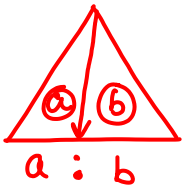
CGL 2017 T2



$ar = \frac{1}{2} a\,b \sin\theta$



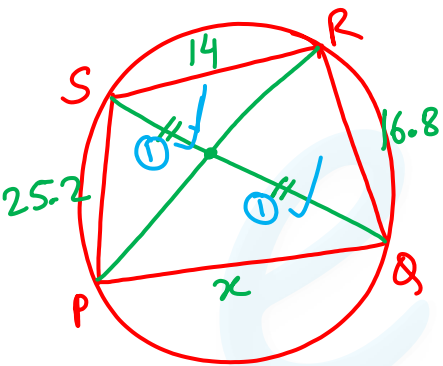




$$\frac{ar\,PSR}{ar\,PQR} = \frac{\frac{1}{2} \times PS \times SR \times \sin\theta}{\frac{1}{2} \times PQ \times QR \times \sin(180-\theta)} = \frac{2(a+b)}{3(a+b)}$$

$$\frac{ar\,PSR}{ar\,PQR} = \frac{ab}{cd} = \frac{2}{3}$$

coaching center



$\frac{14}{21} \times \frac{25.2}{16.8} = \frac{14}{21} \times x$

⑥ $PQRS$ is a cyclic quadrilateral in which $PQ = x\ cm, QR = 16.8\ cm, RS = 14$ cm, $PS = 25.2\ cm$ and PR bisects QS. What is the value of x?

$PQRS$ एक चक्रीय चतुर्भुज है जिसमे $PQ = x cm, QR = 16.8 cm, RS = 14 cm, PS = 25.2 cm$ और PR, QS को सम्द्विभाजित करती है | x का मान क्या है ?

a) 24 b) 21

c) 18 d) 28

$\frac{14 \times 25.2}{x \times 16.8} = \frac{1}{1}$

coaching center

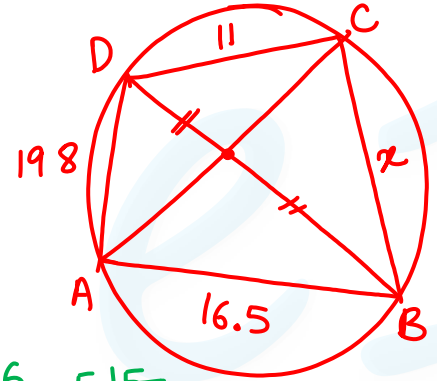
CGL 2019 T1

⑦ $ABCD$ is cyclic quadrilateral in which $AB = 16.5\ cm, BC = x\ cm,$ $CD = 11\ cm,\ AD = 19.8$ cm and BD is bisected by AC at O. What is the value of x?

$ABCD$ एक चक्रीय चतुर्भुज है, जिसमे $AB = 16.5\ cm,\ BC = x\ cm,$ $CD = 11\ cm,\ AD = 19.8\ cm$ और AC, BD को बिंदु O पर दविभाजित करती है| x का मान क्या है ?

a) 12.4 cm b) 13.8 cm

c) 13.2 cm d) 12.8 cm

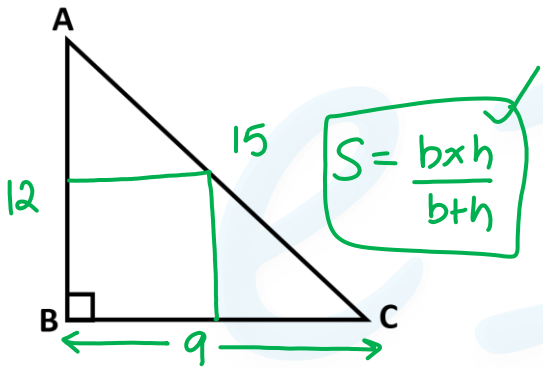


$11 \times 19.8 = 16.5 \times x$

66 5+5

$13.2 = x$

coaching center



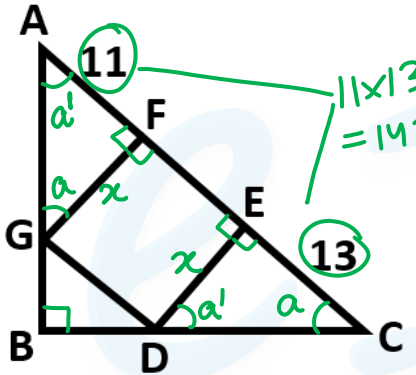
(8) In the given figure, in a right angle triangle $ABC, AB = 12cm$ and $AC = 15\,cm$. A square is inscribed in the triangle. One of the vertices of square coincides with the vertex of triangle. What is the maximum possible area of the square?

दी गई आकृति में एक समकोण त्रिभुज ABC में, $AB = 12cm$ तथा $AC = 15\,cm$ है त्रिभुज के भीतर एक वर्ग बनाया गया है वर्ग के शीर्ष में से एक त्रिभुज के शीर्ष से स्पर्श करता है वर्ग का अधिकतम संभव क्षेत्रफल क्या है?

a) $\frac{1296}{49}$ b) 25

c) $\frac{1225}{36}$ d) $\frac{1225}{64}$

CGL 2017 T-2



$11 \times 13 = 143$

$\frac{11}{x} = \frac{x}{13}$

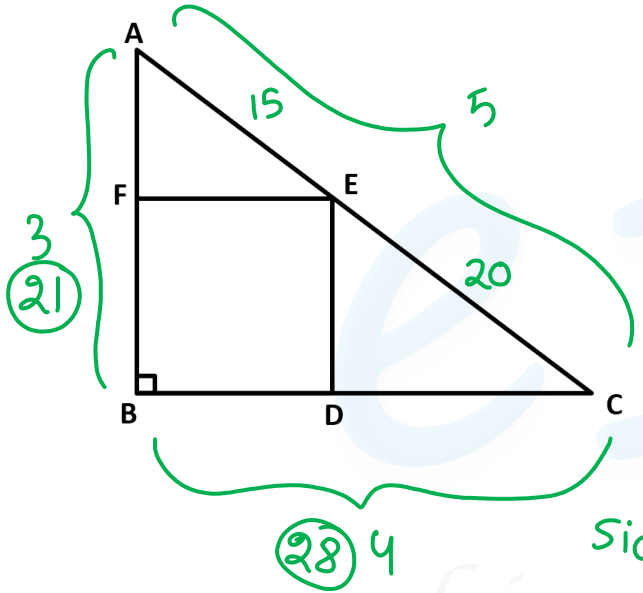
$143 = 11 \times 13 = x^2$

(9) In a right angle triangle ABC, (right angled at B) there are two point on side AC such that $AF = 11\ cm$ and $EC = 13\ cm$. There are two points G and D on sides AB and BC respectively. Find the area of square $DEFG$.

ABC एक समकोण त्रिभुज है जिसमे B समकोण है, इसकी भुजा AC पर दो बिंदु इस प्रकार हैं कि $AF = 11\ cm$ और $EC = 13\ cm$ है| दो बिंदु G और D भुजाओं AB और BC पर स्थित हैं| वर्ग $DEFG$ का क्षेत्रफल पता करो|

a) 121 b) 169
c) 143 d) Can't say

SA



(10) In the adjoining figure ABC is a right angled triangle, BDEF is a square, $AE = 15\ cm$ and $AC = 35\ cm$. What is the area of square BDEF?

दी गई आकृति में ABC एक समकोण त्रिभुज है BDEF एक वर्ग है $AE = 15\ cm$ और $AC = 35\ cm$ है तो वर्ग $BDEF$ का क्षेत्रफल बताए|

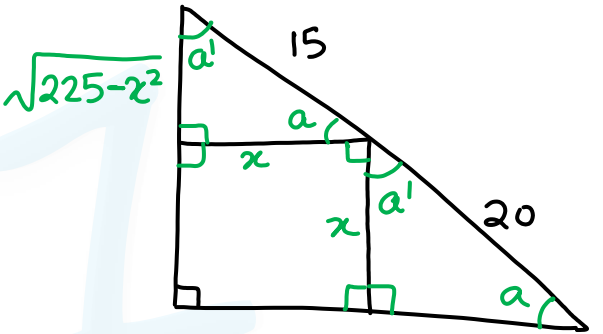
a) $169\ cm^2$ b) $144\ cm^2$

c) $225\ cm^2$ d) $100\ cm^2$

$$\text{side} = \frac{28 \times 21}{49}$$

$$\text{area} = 144$$

coaching center

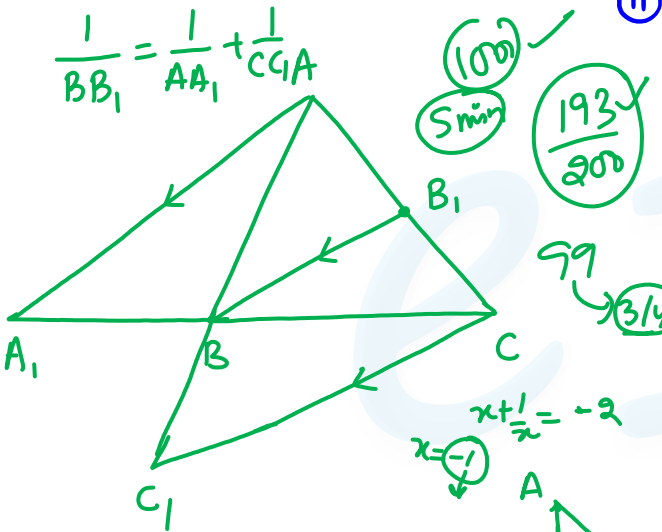


$$\frac{\sqrt{225-x^2}}{x} = \frac{\cancel{15}\ 3}{\cancel{20}\ 4}$$

$$16 \times 225 - 16x^2 = 9x^2$$

$$\Rightarrow 16 \times \overset{9}{\cancel{225}} = \cancel{25}x^2$$

coaching center



(11) B_1 is a point on the side AC of ΔABC and B_1B is joined. A line is drawn through A parallel to B_1B meeting CB at A_1 and another line is drawn through C parallel to B_1B meeting AB produced at C_1. Then

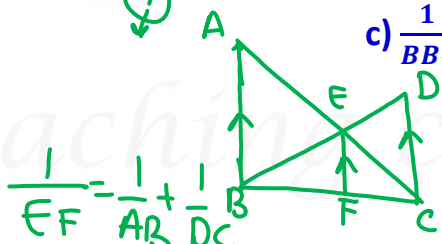
ΔABC की भुजा AC पर B_1 एक बिन्दु है और B_1B के समान्तर A से होकर एक रेखा खींची जाती है जो CB से A_1 पर मिलती है और B_1B के समान्तर एक और रेखा खींची जाती है जो C से होकर जाती है और AB से C_1 पर मिलती है, तो :

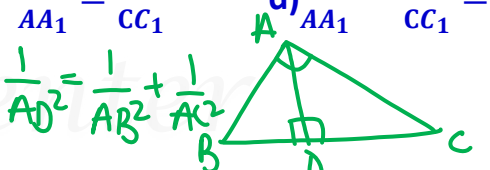
a) $\frac{1}{CC_1} - \frac{1}{AA_1} = \frac{1}{BB_1}$

b) $\frac{1}{CC_1} + \frac{1}{AA_1} = \frac{1}{BB_1}$

c) $\frac{1}{BB_1} - \frac{1}{AA_1} = \frac{2}{CC_1}$

d) $\frac{1}{AA_1} - \frac{1}{CC_1} = \frac{2}{BB_1}$



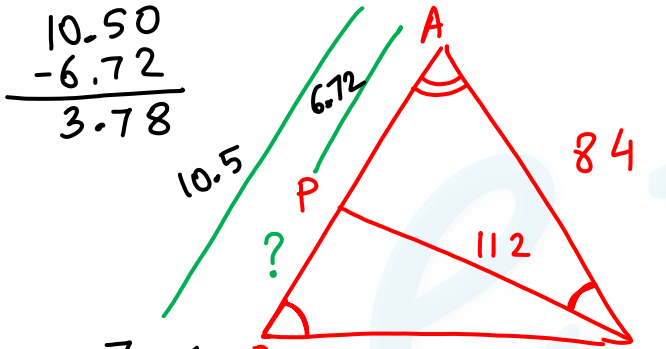


CGL 2016 T-2

(12) In ΔABC, $AC = 8.4\ cm$ and $BC = 14\ cm$. P is a point on AB such that $CP = 11.2\ cm$ and $\angle ACP = \angle B$. What is the length (in cm) of BP?

ΔABC में $AC = 8.4cm$ और $BC = 14\ cm$ है | P, AB पर एक ऐसा बिंदु है जिससे $CP = 11.2\ cm$ और $\angle ACP = \angle B$ होता है | BP की लम्बाई क्या होगी?

a) 3.78 b) 4.12
c) 3.6 d) 2.8



10.50 − 6.72 = 3.78

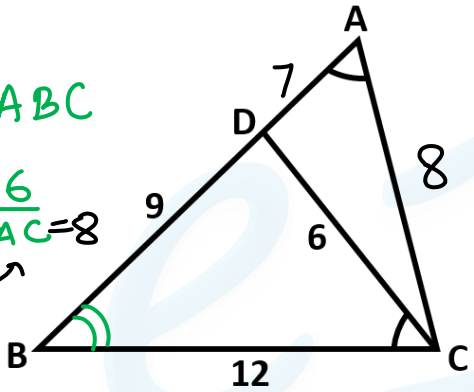
$BA = \frac{14 \times 8.4}{11.2} = 84/2$

$\triangle CAP \sim \triangle BAC$

$\frac{8.4}{BA} = \frac{AP}{8 4} = \frac{11.2}{14}$

$AP = \frac{11.2 \times 8 4}{14} = 6.72$

CGL 2019 T1



CBD ~ ABC

$\frac{12}{AB} = \frac{9}{12} = \frac{6}{AC} = 8$

16

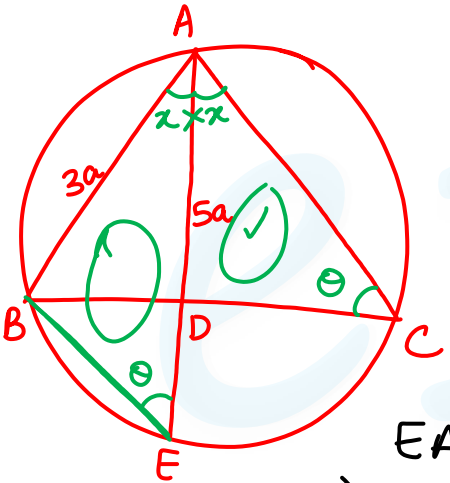
$\frac{ADC}{BDC} = \frac{7}{9}$

(13) **Consider the triangle ABC shown in the following figure where BC = 12cm, DB = 9 cm, CD = 6 cm and $\angle BCD = \angle BAC$. What is the ratio of the perimeter of the triangle ADC to that of the triangle BDC?**

निम्न दिए गए त्रिभुज ABC में BC = 12cm, DB = 9 cm, CD = 6 cm और $\angle BCD = \angle BAC$ है| त्रिभुज ADC और त्रिभुज BDC के परिमापों का अनुपात क्या होगा?

a) $\frac{7}{9}$ b) $\frac{8}{9}$ c) $\frac{6}{9}$ d) $\frac{5}{9}$

CAT 2005



(14) In ΔABC, the angle bisector of $\angle BAC$ cut BC at D and meet the circumcircle of ΔABC at E. If $AB : AD = 3 : 5$ *then find* $AE : AC$?

ΔABC में $\angle BAC$ का द्विभाजक BC को D पर और ΔABC के परिवृत को E पर काटता है यदि $AB : AD = 3 : 5$ तो $AE : AC$ क्या होगा?

a) 2 : 3 b) 3 : 5

c) 5 : 3 d) 3 : 2

$EAB \sim CAD$

$\frac{EA}{CA} = \frac{AB}{AD} = \frac{EB}{CD}$

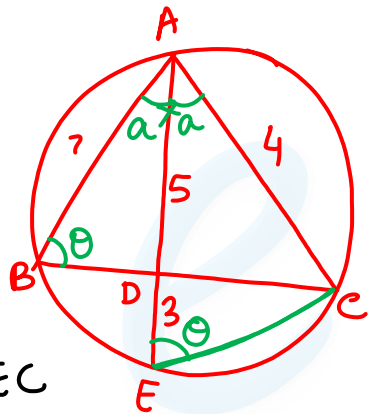
$\frac{3}{5}$

coaching master

(15) In a triangle ABC, angle bisector of $\angle BAC$ cut the side BC at D and meet the circumcircle of ΔABC at E. if $AC = 4\ cm, AD = 5\ cm, DE = 3\ cm$. Find the length of AB.

किसी त्रिभुज ABC में, $\angle BAC$ का सम्दविभाजक भुजा BC को बिंदु D काटता है और ΔABC के परिवृत्त को बिंदु E पर मिलता है| अगर $AC = 4\ cm, AD = 5\ cm, DE = 3\ cm$ है तो AB की लम्बाई पता करो|

a) 10 b) 4 c) 15 d) 8

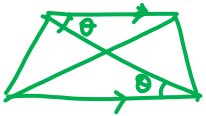


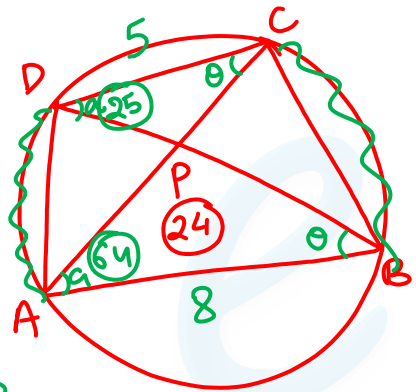
$\overset{\frown}{ABD} \sim AEC$

$\frac{AB}{8} = \frac{5}{4}$

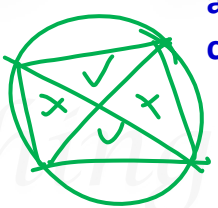
coaching center

SA





$\frac{3}{\cancel{24}} \times 25 = \frac{75}{8}$

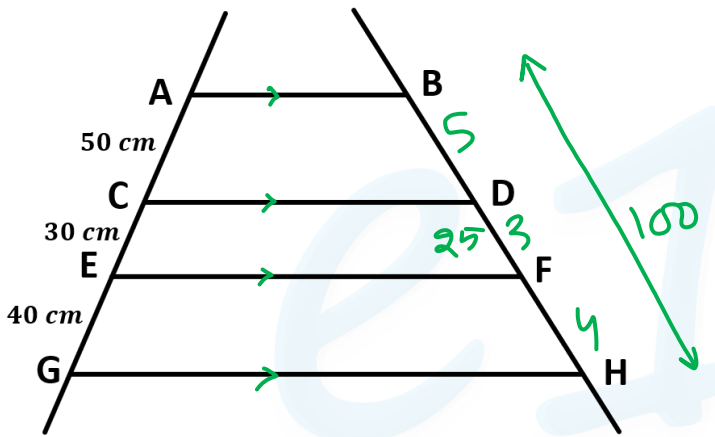


(16) The diagonals of a cyclic quadrilateral $ABCD$ intersect at P and the area of the triangle APB is 24 square cm. If $AB = 8cm$ and $CD = 5cm$, then what is the area of the triangle CPD?

किसी चक्क्रीय चतुर्भुज ABCD के विकर्ण बिन्दु P पर एक दूसरे को काटते हैं और त्रिभुज APB का क्षेत्रफल 24 वर्ग cm है| अगर AB = 8 cm और CD = 5 cm है तो त्रिभुज CPD का क्षेत्रफल क्या होगा?

a) 24 square cm
b) 15 square cm
c) 12.5 square cm
d) 9.375 square cm

CDS 2017 II

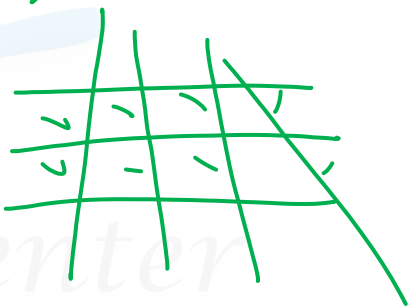


17 In the given figure $AB \parallel CD \parallel EF \parallel GH$ and $BH = 100\ cm$. Find the value of DF:

निम्न चित्र में $AB \parallel CD \parallel EF \parallel GH$ और $BH = 100\ cm$ है| DF का मान:

a) 26 cm b) 40cm

c) 25cm d) 24cm



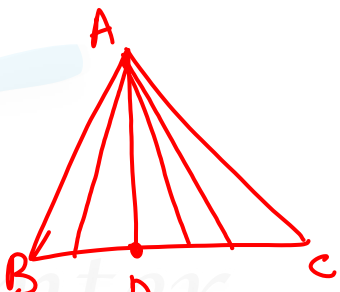
coaching center

Important cevians ?

1. **Median** (माध्यिका)
2. **Altitude** (शीर्ष लम्ब)
3. **Angle bisector** (कोण अर्धक / कोण द्विभाजक)
4. **Perpendicular bisector** (सम्द्विभाजक लम्ब)



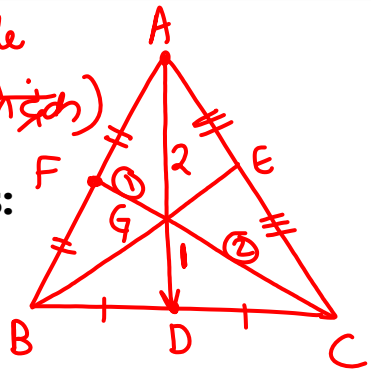
Ceva



coaching center

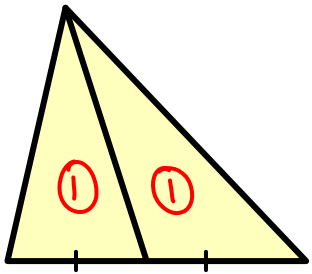
Median (माध्यिका):

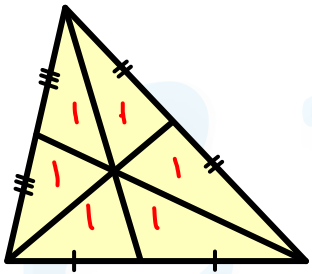
1. **Property:** bisects the opp side
2. **Intersecting point:** Centroid (केन्द्रक)
3. **Ratio property:** 2:1
4. **Location in different triangles:** always inside the △

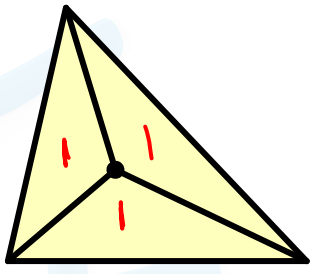


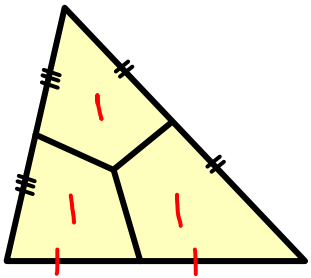
coaching center

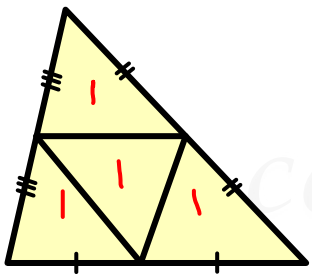
Area division by Medians (माध्यिकाओं द्वारा क्षेत्रफल विभाजन):

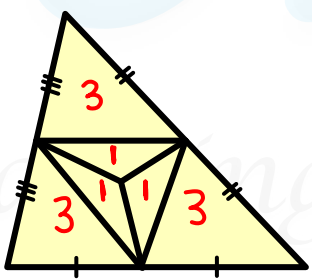


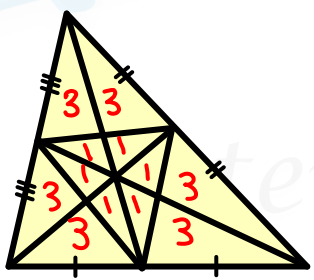


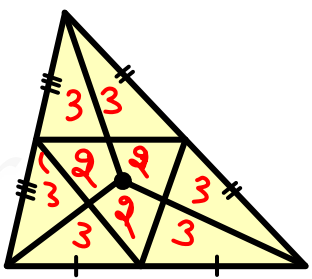




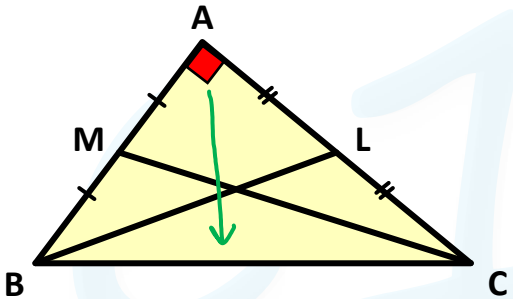




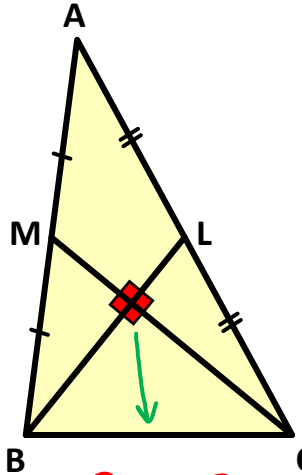




Median special cases:



$5BC^2 = 4(CM^2 + BL^2)$

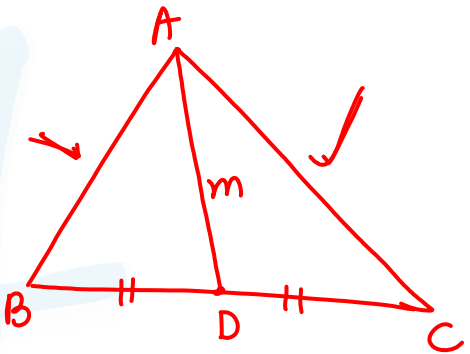


$5BC^2 = AB^2 + AC^2$

coaching centre

Apollonius theorem (अपोलोनियस प्रमेय):

$AB^2 + AC^2 = 2[BD^2 + m^2]$



coaching center

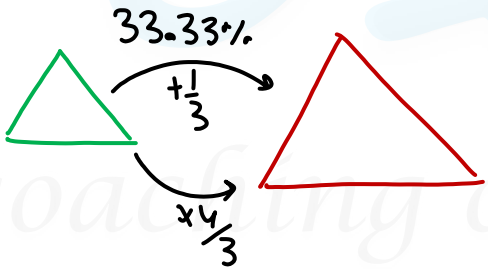
Some standard results:

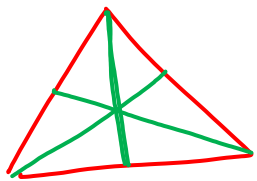


1. Sum of lengths of all medians < sum of lengths of all sides
2. Area of triangle formed by sides is $= \frac{4}{3} \times$ Area of triangle formed by its medians
3. Sum of squares of sides $= \frac{4}{3} \times$ Sum of squares of medians
4. Sum of sides $< \frac{4}{3} \times$ Sum of medians

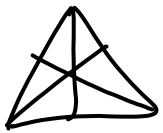
$5^2 = 3^2 + 4^2$

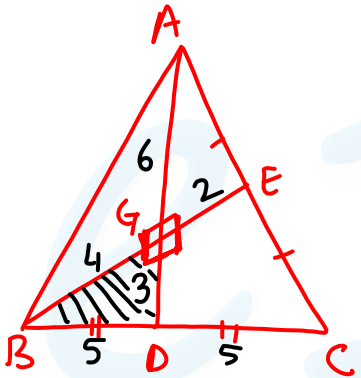
$5 < 3 + 4$





coaching center





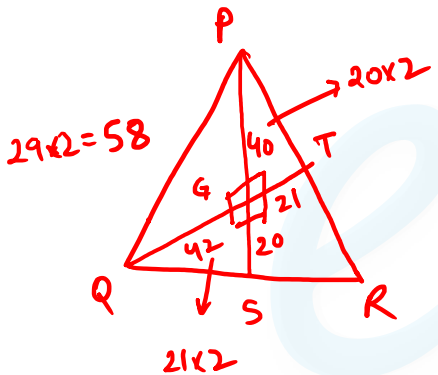
(12) Two medians AD and BE of ΔABC intersect at G at right angle. If $AD = 9\ cm$ and $BE = 6\ cm$, then the length of BC, in cm, is

किसी त्रिभुज ΔABC के दो मध्यिका AD और BE बिंदु G पर समकोण पर मिलते हैं| अगर $AD = 9\ cm$ और $BE = 6\ cm$ है तो BC=?

a) 10 b) 6

c) 5 d) 3

coaching center



20, 21, 29 triplet used

19 HW

G is the centroid of ΔPQR. The medians PS and QT intersect at right angles. If the length of PS and QT are $60cm$ and $63\,cm$ respectively then the length of PQ?

त्रिभुज PQR में G केंद्रक हैं| माध्यिका PS और QT एक दूसरे को समकोण पर काटते है| यदि PS और QT कि लंबाई क्रमश: $60cm$ और $63\,cm$ है| तब PQ लंबाई होगी?

a) 48 b) 58

c) 62 d) 64

coaching center

(20) If G be the centroid of ΔABC such that $AG = BC$, then the magnitude of $\angle BGC$ is

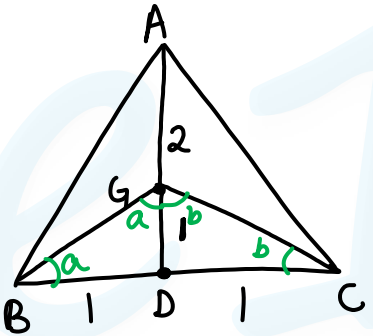
अगर G, त्रिभुज ΔABC का केन्द्रक है तथा $AG = BC$ है तो $\angle BGC$=?

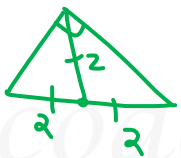
a) 60 ~~b) 90~~

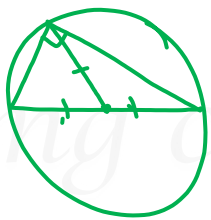
c) 120 d) 135

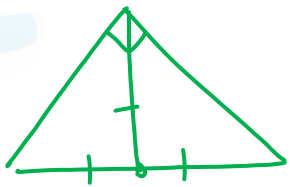
$2a + 2b = 180$

$a + b = 90$

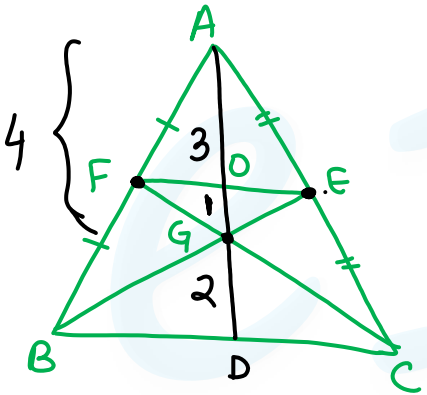








coaching center

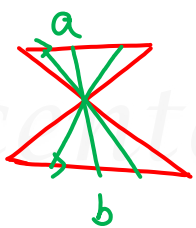


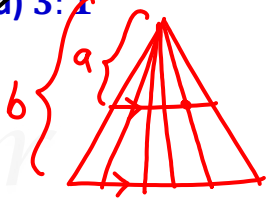
(21) If BE and CF are two median of a ΔABC and G is there intersection point. Similarly O is intersecting point of EF and AG. Find $AO : OG$.

यदि BE तथा CF एक त्रिभुज ABC की दो माधिकाय हों तथा G उनका प्रतिछेद बिन्दु हो ओर EF तथा AG का प्रतिछेद बिन्दु O हो, तो $AO : OG$ कितना होगा?

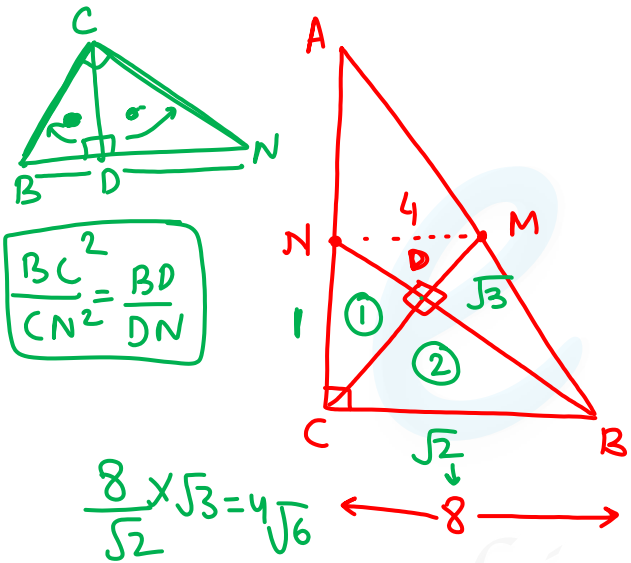
a) 1: 1 b) 1: 2

c) 2: 1 d) 3: 1





coaching content



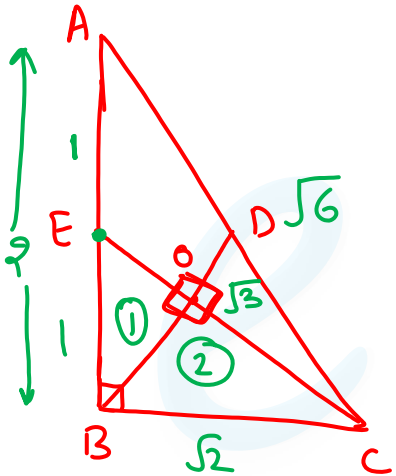
(22) In $\Delta ABC, \angle C = 90°$. M and N are the mid-points of sides AB and AC, respectively. CM and BN intersect each other at D and $\angle BDC = 90°$. If $MN = 4\ cm$, then the length of BN is:

ΔABC में, $\angle C = 90°$ है| M और N क्रमश भुजा AB और AC के मध्य बिंदु हैं| CM और BN एक दूसरे को बिंदु पर प्रतिच्छेद करते हैं और $\angle BDC = 90°$ है| यदि $MN = 4$ सेमी है, तो BN कि लंबाई क्या होगी?

a) $6\sqrt{3}\ cm$ b) $6\sqrt{6}\ cm$

c) $4\sqrt{6}\ cm$ d) $4\sqrt{3}\ cm$

coaching center



(23) In a right angle triangle two medians intersects each other at 90°. Find the ratio of all the three sides of the triangle?

एक समकोण त्रिभुज की कोई दो मध्यिकाएं एक दूसरे को 90° पर काटती है| त्रिभुज कि तीनों भुजाओं का अनुपात ज्ञात करो?

a) $3:4:5$ b) $\sqrt{2}:\sqrt{3}:\sqrt{5}$

c) $1:\sqrt{2}:\sqrt{3}$ d) $\sqrt{3}:2:\sqrt{7}$

$\sqrt{2}:2:\sqrt{6}$

$1:\sqrt{2}:\sqrt{3}$

coaching center